Enumeration by stabilizers of initial strings

Human Compression project: F006 recovery

Working proof; separate review required

Let G be a specified finite group generated by involutions, and let J be its set of nonidentity involutions. A string marking is an ordered tuple $(t_1, \ldots, t_m)$ with $[t_i, t_j] = 1$ for $|i - j| > 1$. We use the usual intersection property (IP): subgroups on two subsets of the marks intersect in the subgroup on their common subset. A C-marking generates its stated group and has IP. Disconnected strings are included. The standard end criterion of McMullen–Schulte, Proposition 2E16, is a retained input: two end deletions with IP and the correct common-middle intersection suffice for a string to have IP.

Lemma 1 (Appending one mark)**.** *Suppose $t = (t_1, \ldots, t_m)$ has IP and $x \in J \setminus \langle t \rangle$ commutes with $t_1, \ldots, t_{m-1}$. There is an exact test for IP of (t, x) using only subgroup generation and intersection. Recursively test the suffix $(t_2, \ldots, t_m, x)$, and require*

$$\langle t_1, \ldots, t_m \rangle \cap \langle t_2, \ldots, t_m, x \rangle = \langle t_2, \ldots, t_m \rangle.$$

The recursion ends with tuples of length at most two, which pass.

Proof. Every prefix encountered during this recursion is a submarking of the old C-marking and therefore has IP. Distinct nonidentity involutions form a rank-two C-marking: their order-two cyclic subgroups have trivial intersection. The hypothesis $x \notin \langle t \rangle$ ensures distinctness in every suffix. Induct on the length, applying the end criterion at each step. Conversely, every suffix and every displayed intersection is required by IP. The recursion is therefore both necessary and sufficient. Its premise that the old prefix has IP is essential; the same abbreviated recursion is not asserted to test arbitrary tuples without that premise. □

Put $A = \mathrm{Aut}(G)$, acting on J. This action is faithful: an automorphism fixing every involution fixes a generating set of G. For each accepted tuple t, let A_t be its pointwise stabilizer in A. The following procedure starts with the empty tuple and trivial subgroup.

1. If $\langle t \rangle = G$, record t and do not extend it.
2. Otherwise form every $x \in J \setminus \langle t \rangle$ commuting with all marks except possibly the last. Take one representative from each A_t-orbit on this candidate set.
3. For each representative passing Lemma 1, visit (t, x) with generated subgroup $\langle t, x \rangle$ and stabilizer $(A_t)_x$.

The candidate set is A_t-invariant, since its defining membership and commutation conditions are invariant under automorphisms fixing t.

Theorem 2. *The procedure terminates and records exactly one representative of each $\mathrm{Aut}(G)$-orbit of generating string C-markings of G. Every possible rank is covered by searching through $\lfloor \log_2 |G| \rfloor$.*

Proof. Each appended mark lies outside the previous subgroup, so the subgroup order grows by an integer index at least two. A visited tuple of length m consequently generates a subgroup of order at least 2^m. This proves the depth bound and termination. A tuple already generating G has no independent extension. Such pruning loses no C-marking, because IP implies that no mark lies in the subgroup on the other marks.

Induct on tuple length. The empty tuple has one orbit. Suppose the visited prefixes represent precisely the orbits of C-markings of length m. Every C-marking of length $m+1$ has one of these prefixes after an automorphism of G. Its final mark is a candidate and passes the exact test. An element of the prefix stabilizer moves it to the chosen orbit representative, so the extended marking is visited. Conversely two extensions can be automorphic only if their prefixes are. Once their prefix representatives agree, that automorphism lies in A_t; distinct chosen candidate orbits therefore cannot merge. This proves coverage and uniqueness, both for partial markings and for generating leaves. □

Lemma 3 (Testing possible completions)**.** *For a visited prefix $t = (t_1, \ldots, t_m)$ put*

$$P_m = J \cap \bigcap_{i=1}^{m-1} C_G(t_i), \qquad K_m = \langle t_1, \ldots, t_m, P_m \rangle.$$

An empty intersection means $P_m = J$. If $K_m \neq G$, the whole branch can be discarded without losing any generating marking. The sets P_m can be maintained by intersection with cached sets $J \cap C_G(t_m)$.

Proof. Every later mark in a string extension has index at least $m + 1$ and therefore commutes with each of $t_1, \ldots, t_{m-1}$. It belongs to P_m, so the completed generated group is contained in K_m. If that group is proper, no completion generates G. For an appended mark x, the required pool is precisely $P_{m+1} = P_m \cap C_G(t_m)$, independently of x. Caching the full commuting set changes neither this equality nor the subgroup test. These pools are invariant under the pointwise prefix stabilizer. Thus the pruning preserves the generating automorphism orbits of Theorem 2. It intentionally changes the counts of visited partial markings: those counts are no longer a census of all partial C-markings. The subgroup test uses every element of the pool, not merely representatives of its stabilizer orbits. □

Lemma 4 (Duality counts)**.** *Let N_m be the number of recorded markings of rank m, and let S_m count those for which an automorphism sends the tuple to its reverse. There are $(N_m + S_m)/2$ classes up to isomorphism and duality, of which S_m are self-dual. The same formula holds after restricting to tuples whose adjacent products all have order greater than two.*

Proof. Reversal induces an involution on the finite set of ordered automorphism orbits. Its fixed points are exactly the S_m specified orbits, and its other orbits have size two. The connected-string condition is preserved by automorphisms and reversal. Finally, the correspondence between regular polytopes and string C-groups identifies an isomorphism of two markings on the same group with an automorphism sending their distinguished generators in order. Reversal is precisely duality in that correspondence. □

Implementation and retained inputs. The complete GAP program implements these statements literally. One version forms J from all group elements; the general version unions all conjugacy classes whose representatives have order two. It checks that their union has the sum of their class sizes. It computes the full automorphism group and its faithful permutation action on J, checks faithfulness by exact orders, and uses point stabilizers and candidate orbits. The recursive test is called only on extensions of already accepted prefixes. Complete leaf tuples, automorphism generators, involutions and counts at every rank are retained, including zero counts above the last surviving prefix.

The groups are supplied by exact reflection permutations on finite root sets. The classical finite Coxeter presentations and orders identify these models: the simple reflections satisfy the specified relations, and their generated permutation groups have the classical orders. The complete root constructor, permutations, group computations and run logs are required finite evidence, not optional validation. GAP's exact group operations, complete conjugacy and automorphism algorithms, and their actual native execution are explicit software premises. This is not a Lean or independently kernel-checked computation. A timeout is never a classification certificate. The variant with cached commuting sets and the completion-subgroup test uses the additional proved lemma above. Its visited-prefix counts are reported with their pruning counts; only complete generating leaves and their duality classes are used in the source comparison.

Chen–Rowley, *Abstract Regular Polytopes of Finite Irreducible Coxeter Groups*, EJC 32(4) (2025), P4.18, supplies the target Table 1. The stabilizer enumeration is the established method described by Leemans–Mixer, *Algorithms for classifying regular polytopes with a fixed automorphism group*, Contributions to Discrete Mathematics 7(2) (2012), 105–118, Section 5, there attributed to Hartley–Hulpke, *Polytopes derived from sporadic simple groups*, Contributions to Discrete Mathematics 5(2) (2010), 106–118. The proof here supplies the precise extension test and completion-subgroup pruning for our executable version. It uses ordinary group actions and the retained end criterion; no priority for these general methods is asserted. A separate source comparison must distinguish connected and disconnected markings and record every completed exceptional row. Until those executions and their review are complete, this note claims no whole-source recovery or numerical compression factor.